

Unbiased estimators

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{where } x_i \sim N(\mu, \sigma^2)$$

$$E(\hat{\mu}) = E\left(\frac{1}{n} \sum_{i=1}^n x_i\right) = \frac{1}{n} \sum_{i=1}^n E(x_i) = \mu \quad \therefore \text{unbiased}$$

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{where } x_i \sim \text{Bernoulli}(p)$$

$$E(\hat{p}) = \frac{1}{n} \sum_{i=1}^n E(x_i) = p \quad \therefore \text{unbiased}$$

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$$X_1, \dots, X_n \sim \text{Unif}(0, \theta)$$

$$(1) \hat{d}_1(\theta) = 2 \sum_{i=1}^n x_i / n \quad E(x_i) = \frac{\theta}{2}$$

$$(2) \hat{d}_2(\theta) = \max_i \{x_i\}_{i=1}^n$$

$$E(\hat{d}_1(\theta)) = \frac{2}{n} \sum_{i=1}^n E(x_i) = \frac{2}{n} \left(\frac{\theta}{2}\right)n = \theta \quad \therefore \text{unbiased}$$

$$E(\hat{d}_2(\theta)) = ?$$

$$P(\hat{d}_2(\theta) \leq x) = P(\max_i x_i \leq x)$$

$$\begin{aligned} &= P(x_1 \leq x, x_2 \leq x, \dots, x_n \leq x) \\ &= \prod_{i=1}^n P(x_i \leq x) = (x/\theta)^n \end{aligned}$$

$$f_{d_2}(x) = \begin{cases} \frac{nx^{n-1}}{\theta^n} & \text{for } 0 \leq x \leq \theta \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned} E(\hat{d}_2) &= \int_0^\theta f_{d_2}(x) x \, dx \\ &= \int_0^\theta \frac{nx^{n-1}}{\theta^n} x \, dx = \frac{n}{\theta^n} \int_0^\theta x^n \, dx \\ &= \frac{n}{\theta^n} \left(\frac{x^{n+1}}{n+1} \right)_0^\theta = \frac{n}{n+1} \frac{\theta^{n+1}}{\theta^n} \\ &= \frac{n\theta}{n+1} \neq \theta \end{aligned}$$

estimator with high variance

$$X_1, \dots, X_n \sim N(\mu, \sigma^2) \quad \left[\begin{array}{l} \text{could be any} \\ \text{other dist.} \end{array} \right]$$

$$\hat{\mu} = \bar{X}_n \quad \text{unbiased as } E(\hat{\mu}) = \mu$$

$$\text{Var}(\hat{\mu}) = \sigma^2/n$$

$$\text{Estimator MSE} = E[(\hat{\theta} - \theta)^2]$$

We want to prove $\text{MSE} = \text{Var} + \text{bias}^2$

$$= E[(\hat{\theta} - E(\hat{\theta}))^2] + (E(\hat{\theta}) - \theta)^2$$

$$\text{LHS} = E\left[\underbrace{(\hat{\theta} - E(\hat{\theta}))}_{\text{deviation from mean}} + \underbrace{(E(\hat{\theta}) - \theta)}_{\text{bias}}\right]^2$$

$$= E \left[\underbrace{(\hat{\theta} - E(\hat{\theta}))^2}_{\text{Variance}} + (E(\hat{\theta}) - \theta)^2 + 2(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta) \right]$$

$$= E \left[\underbrace{(\hat{\theta} - E(\hat{\theta}))^2}_{\text{Var.}} \right] + E \left[(E(\hat{\theta}) - \theta)^2 \right] + 2 E \left[(\hat{\theta} - E(\hat{\theta}))(E(\hat{\theta}) - \theta) \right]$$

$$= \text{var} + (E(\hat{\theta}) - \theta)^2$$

$$+ 2(E(\hat{\theta}) - \theta) E[(\hat{\theta} - E(\hat{\theta}))]$$

$$= \text{var.} + \text{bias}^2 + 2(E(\hat{\theta}) - \theta) \underbrace{(E(\hat{\theta}) - E(E(\hat{\theta})))}_0$$

$$\therefore E(\hat{\theta}) = E[E(\hat{\theta})]$$

$$\text{Var}(\hat{d}_2) = E(\hat{d}_2^2) - \underbrace{(E(\hat{d}_2))^2}$$

$$E(\hat{d}_2^2) = \int_0^{\theta} x^2 \frac{n x^{n-1}}{\theta^n} dx = \frac{n}{\theta^n} \int_0^{\theta} x^{n+1} dx$$

$$= \frac{n}{\theta^n} \left(\frac{x^{n+2}}{n+2} \right)_0^{\theta} = \frac{\theta^2 n}{n+2}$$

$$\text{Var}(\hat{d}_2) = \frac{\theta^2 n}{n+2} - \frac{n^2 \theta^2}{(n+1)^2} = \frac{n \theta^2}{(n+2)(n+1)^2}$$

$$\text{MSE} = \text{Var} + \text{bias}^2$$

$$= \frac{n \theta^2}{(n+2)(n+1)^2} + \left(\theta - \frac{n \theta}{n+1} \right)^2$$

$$= \frac{n \theta^2}{(n+2)(n+1)^2} + \theta^2 \left(\frac{n+1 - n}{n+1} \right)^2 \rightarrow 0 \text{ as } n \rightarrow \infty$$